Digital Objects in Research-Based Teaching

Roman Coins, Personifications and Orange Datamining

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Part I

Personifications on Roman coins

RRC 415/1
62 BC

RIC IV Philip I 125
AD 246–248

RIC II/1^2 Vespasian 821
AD 75

RIC II/3^2 Hadrian 2576
AD 136–138

Personifications in ancient rhetorical discourse

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Rhet. Her. **4.66.1 (trans. Caplan)**

1. Conformatio est, cum aliqua, quae non adest, persona confingitur quasi adsit, aut cum res muta aut informis fit eloquens, et forma ei et oratio adtribuitur ad dignitatem adcommodata, aut actio quaedam, hoc pacto:

“Quodsi nunc haec urbs invictissima vocem mittat, non hoc pacto loquatur: ‘Ego illa plurimis tropeis ornata, triumphis ditata certissimis, clarissimis locupletata victoriis, nunc vestris seditionibus, o cives, vexor; quam dolis malitiosa Kartago, viribus probata Numantia, disciplinis erudita Corinthus labefactare non potuit, eam patimini nunc ab homunculis deterrumis proteri atque conculcari?’”

Personification consists in representing an absent person as present, or in making a mute thing or one lacking form articulate, and attributing to it a definite form and a language or a certain behaviour appropriate to its character, as follows:

“But if this invincible city would now give utterance to her voice, would she not speak as follows? ‘I, city of renown, who have been adorned with numerous trophies, enriched with unconditional triumphs, and made opulent by famous victories, am now vexed, O citizens, by your dissensions. Her whom Carthage with her wicked guile, Numantia with her tested strength, and Corinth with her polished culture, could not shake, do you now suffer to be trod upon and trampled underfoot by worthless weaklings?’”

Part II

Pattern recognition

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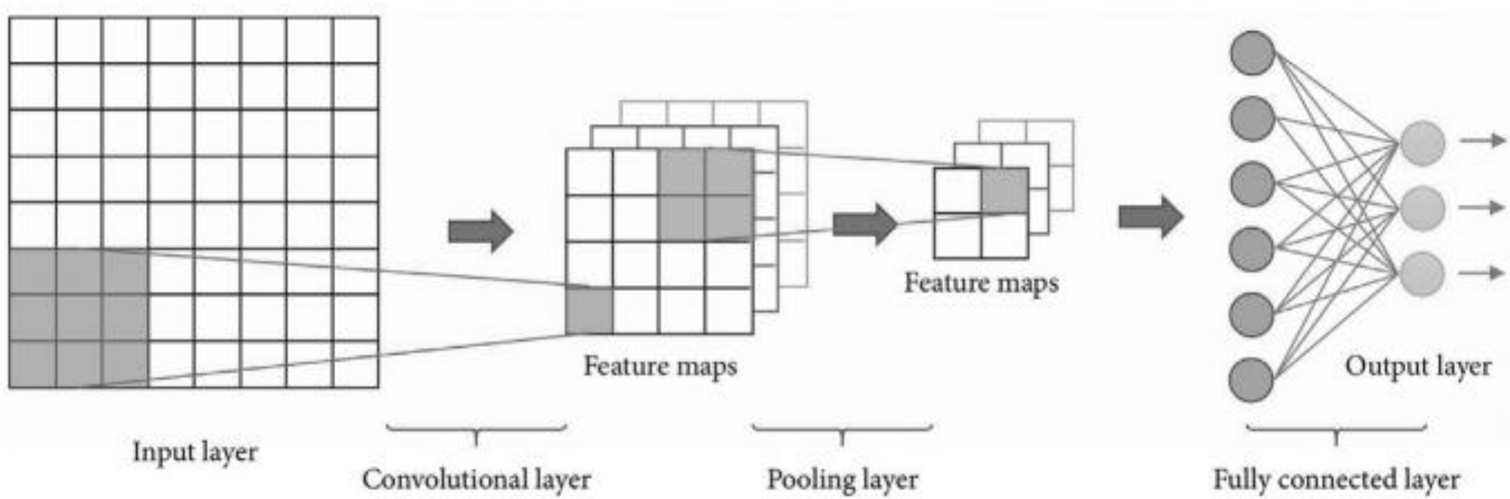
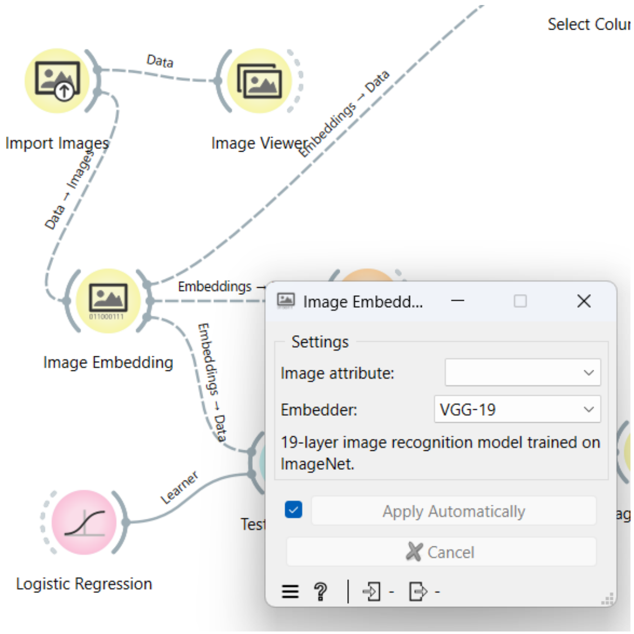


Fig. 1: Pattern recognition with an artificial neural network (from Meruane et al. 2021, fig. 1).

Image source: https://doi. org/10.1155/2021/1483594

Orange Datamining and AI



- SqueezeNet: Small and fast model for image recognition trained on ImageNet.
- Inception v3: Google's Inception v3 model trained on ImageNet.
- VGG-16: 16-layer image recognition model trained on ImageNet.
- VGG-19: 19-layer image recognition model trained on ImageNet.
- Painters: A model trained to predict painters from artwork images.
- DeepLoc: A model trained to analyze yeast cell images.

https://orangedatamining.com/widget-catalog/image-analytics/imageembedding/

Orange Datamining and AI

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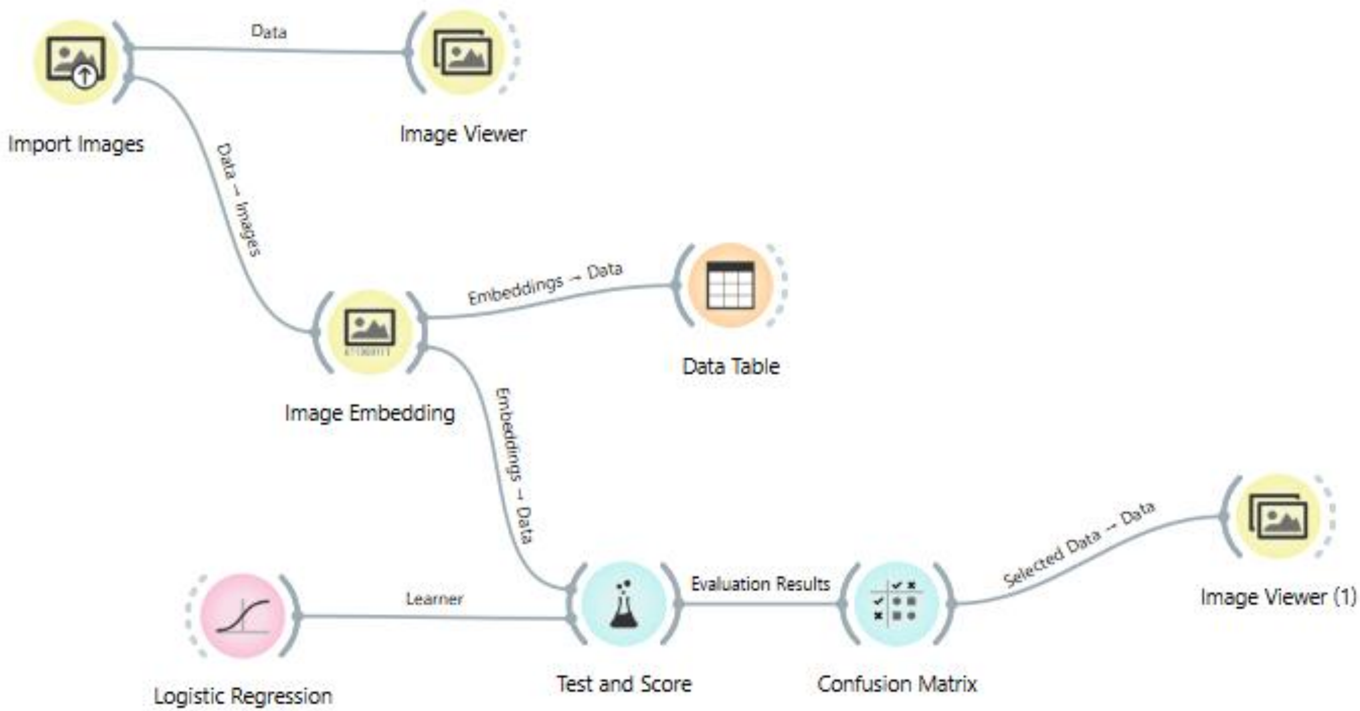
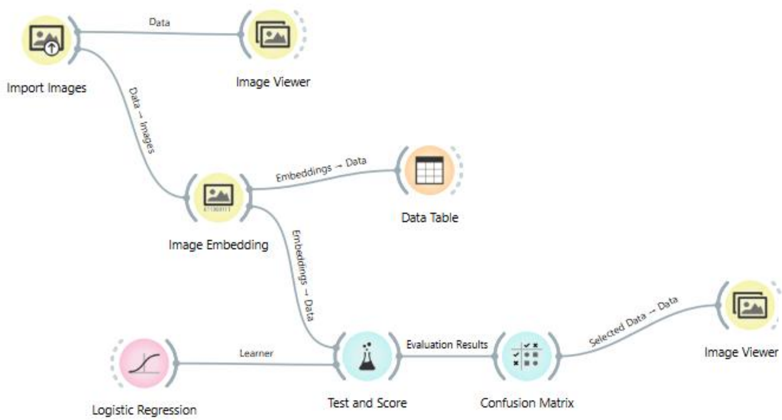
What are limits of “Orange” when we use it to identify personifications on Roman coins?

Confusion Matrix



Confusion Matrix



Predicted

Actual	Aequitas	Concordia	Venus	Σ
Aequitas	5	11	6	22
Concordia	8	12	4	24
Venus	6	3	13	22
Σ	19	26	23	68

VGG 19; AUC = 0,673

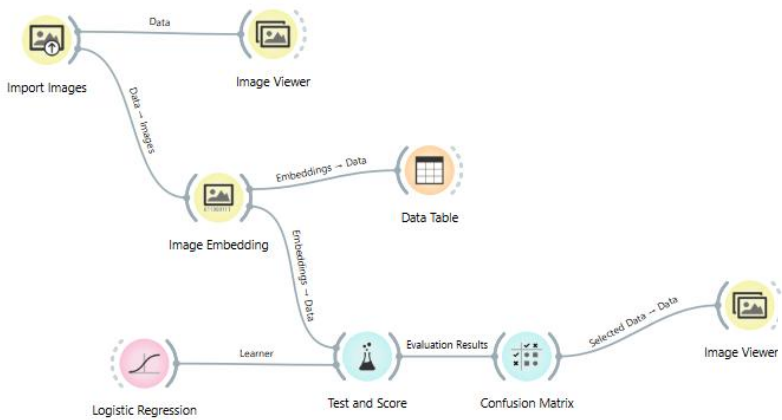
Predicted

Actual	Aequitas	Concordia	Venus	Σ
Aequitas	11	6	5	22
Concordia	3	15	6	24
Venus	4	5	13	22
Σ	18	26	24	68

SqueezeNet; AUC = 0,742

https://orangedatamining.com/widget-catalog/evaluate/testandscore/

Confusion Matrix



Predicted

Actual		Aequitas	Concordia	Venus	Σ
	Aequitas	5	11	6	**22**
	Concordia	8	12	4	**24**
	Venus	6	3	13	**22**
	Σ	**19**	**26**	**23**	**68**

VGG 19; AUC = 0,673

Predicted

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SqueezeNet; AUC = 0,742

To what extent can the Confusion Matrix / AUC tell us how well the pattern recognition works?
Where are the limits of these methods?

Confusion Matrix

Coins correctly classified

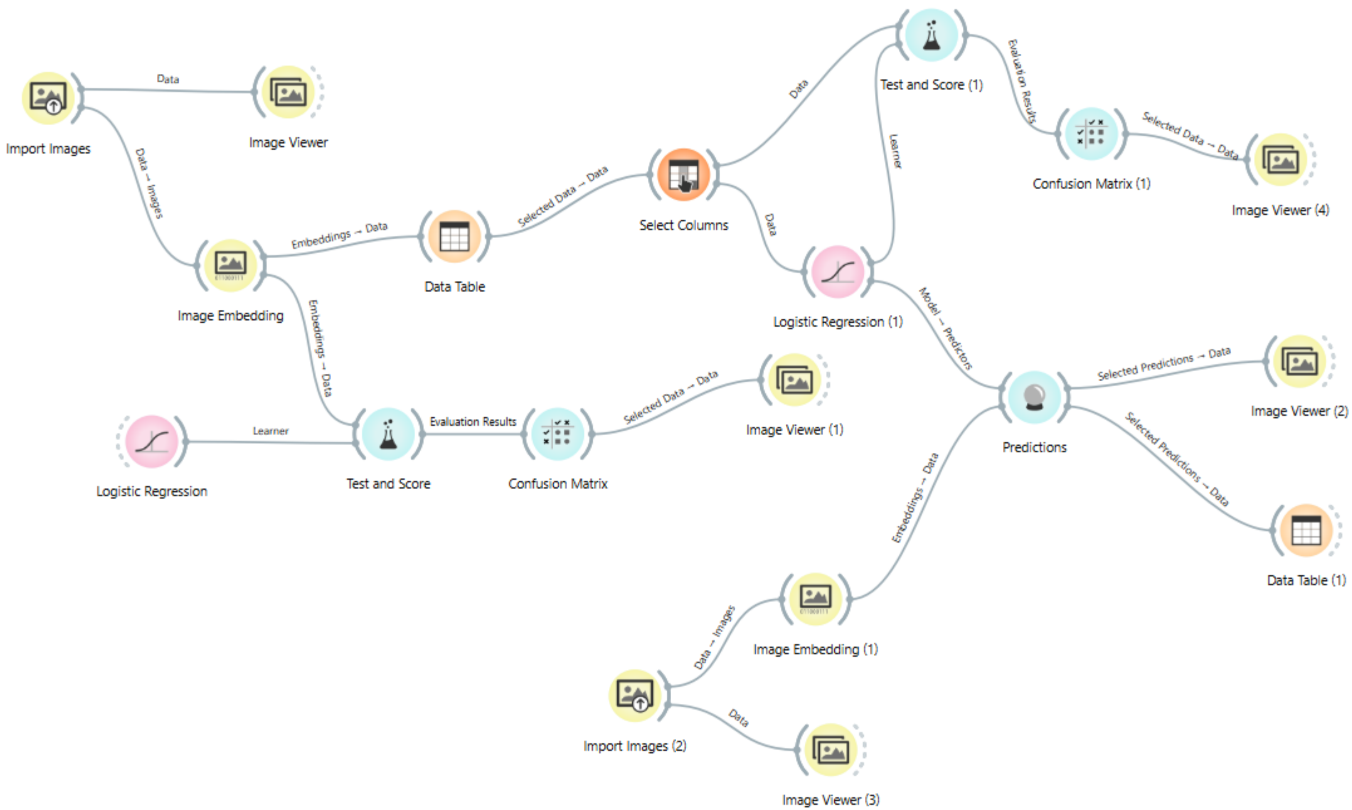
Can we reconstruct which patterns determine the classification?

Confusion Matrix

Coins misclassified (predicted to be Venus)

Can we reconstruct which patterns determine the classification?

A machine to identify personifications?



A possible feature to use for coin attribution?

A machine to identify personifications?

IHACOINS-00001_r IHACOINS-00080_r RICII-1_2nded-Vespasin-821

	Logistic Regression (1)	image name	
1	0.00 : 0.00 : 1.00 → Venus	IHACOINS-00001_r	IHA
2	0.00 : 0.99 : 0.01 → Concordia	IHACOINS-00080_r	IHA
3	0.70 : 0.28 : 0.02 → Aequitas	RICII-1_2nded-Vespasin-821	RICI

A possible feature to use for coin attribution?

A machine to identify personifications?

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Show probabilities for: Classes known to the model

	Logistic Regression (1)	image name	
1	0.00 : 0.00 : 1.00 → Venus	IHACOINS-00001_r	IHAC
2	0.04 : 0.95 : 0.01 → Concordia	IHACOINS-00010_r	IHAC
3	0.00 : 0.99 : 0.01 → Concordia	IHACOINS-00080_r	IHAC

A possible feature to use for coin attribution?

A machine to identify personifications?

Show probabilities for Classes known to the model

	Logistic Regression (1)	image name	
1	0.00 : 0.00 : 1.00 → Venus	IHACOINS-00001_r	IHAC
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3	0.00 : 0.99 : 0.01 → Concordia	IHACOINS-00080_r	IHAC

A possible feature to use for coin attribution?

Part III

Exercise:

Analyse the image grids on the following slide.

The **Image Grid** widget can display images from a dataset in a similarity grid - images with similar content are placed closer to each other. It can be used for image comparison, while looking for similarities or discrepancies between selected data instances (e.g. bacterial growth or bitmap representations of handwriting).
https://orangedatamining.com/widget-catalog/image-analytics/imagegrid/

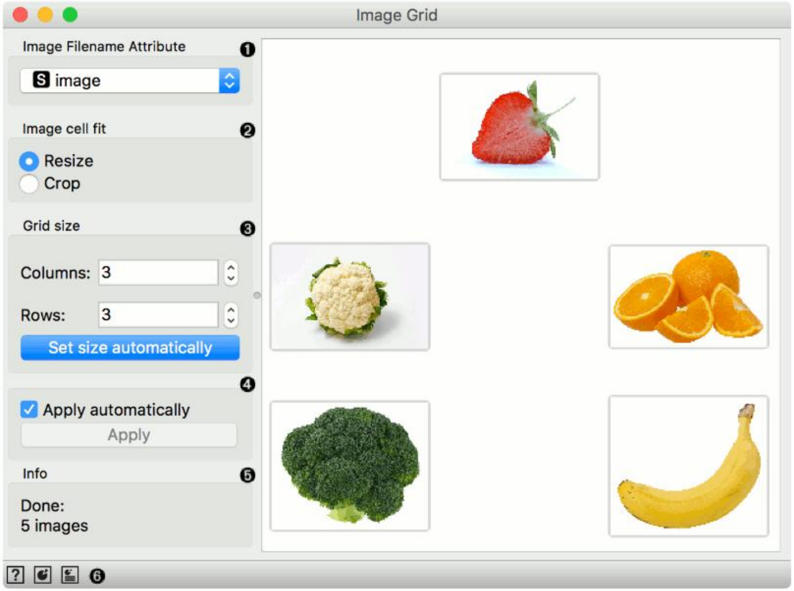

Image Grid: photos of fruits
(see Orange Datamining link above)

Exercise:

1. Evaluate: Which patterns in the images are significant for the estimated „similarity“?
2. Compare the two grids: Can you figure out differences between the two embedders (SqueezeNet vs. VGG 19)? Why are there differences?
3. Discuss: What are differences between the „similarity“ of image patterns and „similarity“ cosidered by human numismatists?

Concordia, SqueezeNet

Concordia, VGG 19

Exercise:

Concordia, SqueezeNet

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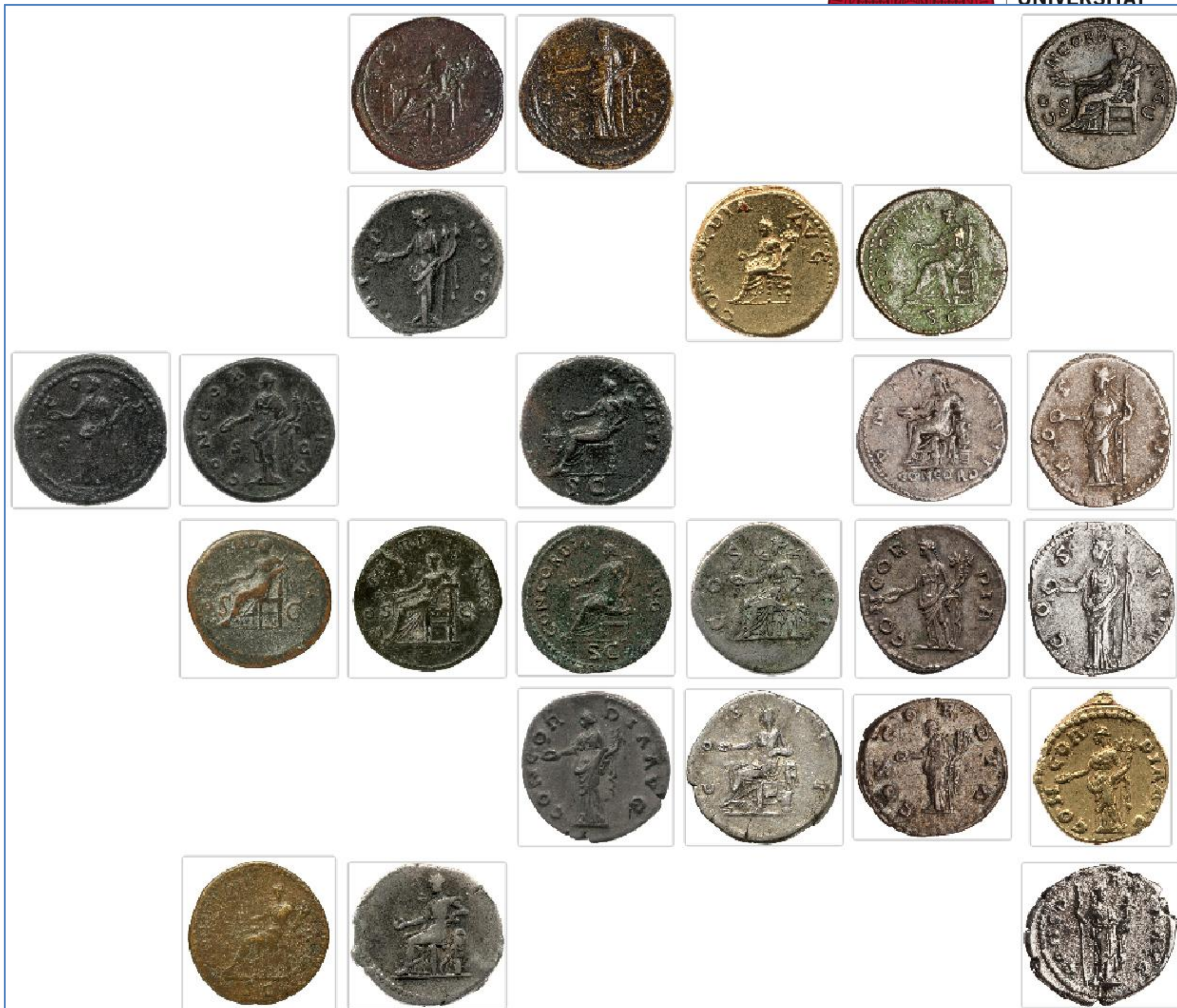
Exercise:

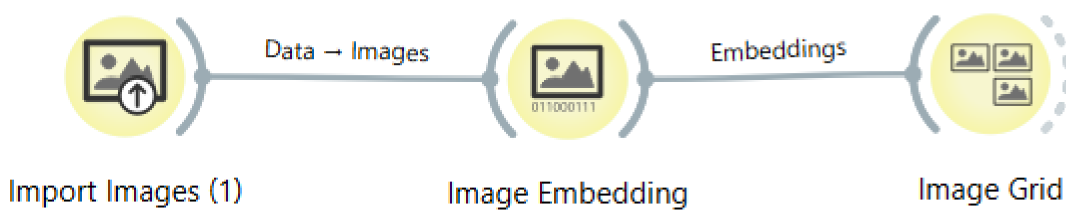


Concordia, VGG 19

Exercise [additional]:

For those of you who would like to try out Orange Datamining themselves, here comes a short guide:

1. Download Orange Datamining, https://orangedatamining.com/download/
2. Add the widgets „Image Analytics“: Choose „Options“ – „Add Ons“ – check box of „Image Analytics“
3. Open a new file, and choose from the left side bar under „Image Analytics“ the following widgets: „Import Images“, „Image Embedding“, „Image Grid“ (see below)
4. Click on one widget and draw a line to the next one to produce a data pipeline.



Exercise [additional]:

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1. Import Images: Download the dataset here: https://heibox.uni-heidelberg.de/d/b1f090de1dc744918347/
2. Doubleclick on „Import Images“ and chose the downloaded file „Concordia“
3. Doubleclick on “Image Embedding” and choose an embedder (in our example “SqueezeNet” and “VGG 19”)
4. Click on “Image Grid” to see the resulting grids.
5. You can now experiment with the different embedders and different personifications.

Conclusion

What do you think:

How would we need to improve the pattern recognition for a more reliable tool with regard to classifying personifications on Roman coins?

Do you think pattern recognition is in general a useful tool to explore personifications on Roman coins?